Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: **Bachelor of Science in Electrical Engineering (BSEE)**

Primary Job Roles students are prepared for in the program	• **Power Engineer** • **Design Engineer** ***(Design and preparation of plans, specifications and estimates for electric power systems, power plants, power distribution systems including power transformers, transmission lines and network protection, switchgear, building wiring, electrical machines, equipment and others)***
Secondary Job Roles students are prepared for in the program	**Illumination Engineer · Entrepreneur · Sales Engineer · Distribution Engineer · Engineering Educators and Researcher · Instrumentation and Control Engineer · Safety Engineer · Maintenance Engineer · Construction and Project Engineer · Software Developer · Electrical Design Inspector**

The scope of practice of Electrical Engineering is defined in Section 2a of the prevailing Electrical Engineering Law or RA 7920 and pertains to professional services and expertise including, but not limited to the job roles stated above.

Current CMO Guiding Program: CHED CMO 88, Series of 2017

In accordance with the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of an outcomes-based quality assurance system as advocated under CMO 46 s. 2012 (Policy-Standard to Enhance Quality Assurance (QA) in Philippine Higher Education through an Outcomes-Based and Typology-Based Quality Assurance) and as addendum to CMO 37, s. of 2012 (Establishment of an Outcomes-Based Educational System in Higher Education Institutions offering Engineering Programs), and by virtue of the Commission en banc through Resolution No. 788-2017 dated October 24, 2017 the following Policies, Standards and Guidelines (PSG) are hereby adopted and promulgated by the Commission.

Program Description:

Degree name

The degree program described herein shall be called Bachelor of Science in Electrical Engineering (BSEE).

Nature of the Field of Study

The field of Electrical Engineering deals with the generation, transmission, distribution and utilization of electricity. It also deals with the design, operation and protection, maintenance and economics of electrical systems with emphasis on ethical values to harness economically and safely the materials, and forces of nature for the benefit of society and the environment.

The BSEE curriculum is designed to develop engineers who have a background in mathematics, natural, physical and allied sciences. As such, the curriculum contains courses in mathematics, science and engineering fundamentals with emphasis on the development of analytical and creative abilities. It also contains language courses, social sciences and humanities. This is to ensure that the electrical engineering graduate is articulate and is able to understand the nature of his/her special role in society and the impact of his/her work on the progress of electricalization.

The curriculum is designed to guarantee a certain breadth of knowledge of the Electrical Engineering disciplines through a set of core courses. It ensures depth and focus in certain disciplines through areas of specialization. It provides a recommended track of electives that HEIs may adopt or develop. The curriculum develops the basic engineering tools necessary to solve problems in the field of Electrical Engineering. This enables the graduate to achieve success in a wide range of career.

Institutional electives are prescribed in order to give a certain degree of specialization so that institutions of learning will develop strengths in areas where they already have a certain degree of expertise.

Emphasis is given to the basic concepts. Previously identified courses are strengthened to take into account new developments. New courses and/or topics are introduced so that the student's knowledge of the fundamentals may be enhanced. This is to allow the student to achieve a degree of knowledge compatible with international standards.

VISION

A PREMIER UNIVERSITY IN THE ASEAN REGION
By becoming a premier university in the ASEAN Region, the USEP shall be a center of excellence and development, responsive and adaptive to fast-changing environments. USEP shall be known as the leading university in the country that fosters innovation and applies knowledge to create value towards social, economic, and technological developments.

MISSION

USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management.
Particularly, USEP is committed to:
1. Provide quality education for students to grow in knowledge, promote their well-rounded development, and make them globally competitive in the world of work;
2. Engage in high impact research, not only for knowledge's sake, but also for its practical benefits to society; and,
3. Promote entrepreneurship and industry collaboration

CORE VALUES

USeP is a community of scholars that values:
C - Collaboration
A - Accountability
R - Resilience
E - Excellence
S - Service Centered

Program Educational Objectives		Mission-Vision and Core Value Alignment								Reference Source or CMO Provision Compliance
		Vision	Mission			Core Value				
			1	2	3	U	S	E	P	
PEO1	Demonstrate expertise in the field of Electrical Engineering	√	√	√	√		√	√	√	USEP Brochure, Course Syllabus, CMO 88
PEO2	Actively participate in professional communities to nourish ever-developing careers	√	√	√	√	√	√		√	USEP Brochure, Course Syllabus, CMO 88
PEO3	Demonstrate leadership and commitment to work in diverse cultural and interdisciplinary backgrounds	√	√	√	√	√		√	√	USEP Brochure, Course Syllabus, CMO 88
PEO4	Exhibit globally-competitive skills	√	√	√	√		√	√	√	USEP Brochure, Course Syllabus, CMO 88

PERFORMANCE INDICATORS

Institutional Graduate Attributes:

IGA1 – Critical Analysis
IGA2 – Leadership and Teamwork
IGA3 – Service-Oriented
IGA4 – Professional Competence
IGA5 – Communication
IGA6 – Life-Long Learning

Program Attribute	Program Attribute (New)		Program Outcomes		Performance Indicators
IGA1	IGA2	a	Apply knowledge of mathematics and sciences to solve engineering problems;	1	Apply concepts in mathematics to solve engineering problems
IGA4	IGA5			2	Apply principles in chemical sciences to solve engineering problems.
IGA6	IGA4			3	Apply principles in physical sciences to solve engineering problems
IGA1, IGA4	IGA2, IGA5	b	Design and conduct experiments, as well as to analyze and interpret data	1	Perform laboratory experiments as prescribed by standard procedures.
IGA5, IGA6	IGA4, IGA5			2	Design an experiment to achieve a specific objective
IGA1, IGA2	IGA1, IGA 2	c	Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	1	Develop an engineering solution that have various constraints as factors
IGA3, IGA4	IGA3, IGA4, IGA5			2	Design a system or process based on various constraints
IGA5, IGA6					
IGA2	IGA1			1	Shows tact in dealing with people he/she comes in contact with.

IGA3	IGA3	**d**	Function on multidisciplinary teams	2	Capable of learning from and listening to co-members

IGA6	IGA4			3	Shows respect and courtesy in dealing with peers and superiors.
IGA1	IGA2	**e**	Identify, formulate, and solve engineering problems	1	Interpret statement problems into mathematical expressions.
IGA4	IGA5			2	Apply concepts, theories, and principles in solving engineering problem(s)
IGA6	IGA4			3	Solve specific engineering problem related to discipline.
IGA4	IGA5	**f**	Understand professional, social and ethical responsibility	1	Internalizes the professional code of ethics for specific discipline.
IGA5	IGA4, IGA5			2	Applies the principles of professional ethical standard and social responsibility to actual practice
IGA6					
IGA4	IGA5	**g**	Communicate effectively	1	Articulate the convention and style in oral communication that conveys complete thought and information.
IGA5	IGA5			2	Articulate the convention and style in written communication that conveys complete thought and information.
IGA6	IGA4			3	Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
IGA1, IGA3	IGA2, IGA3	**h**	Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education	1	Relate engineering solutions to global, economic, environmental, and societal context.
IGA4, IGA6	IGA4, IGA5			2	Describe engineering solutions in application to global, economic, environmental, and societal context.
IGA1, IGA2	IGA1, IGA2	**i**	Recognize the need for, and an ability to engage in life-long learning	1	Demonstrate ability to learn independently.
IGA3, IGA4	IGA3, IGA5			2	Shows appreciation and understanding of the engineering profession.
IGA5, IGA6	IGA4, IGA5			3	Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
IGA2, IGA3	IGA1, IGA3	**j**	Apply knowledge of contemporary issues	1	Identify the socio-economic, political, and environmental issues related to the practice of engineering
IGA4, IGA5	IGA5			2	Explain the possible impact of technology to local, national, and international issues
IGA6	IGA4			3	Shows leadership skills in managing a multidisciplinary group.
IGA1	IGA2	**k**	Use techniques, skills, and modern engineering tools necessary for engineering practice	1	Handles the details of the work assigned to him/her.
IGA4	IGA5			2	Demonstrate the ability to operate the equipment needed in the engineering practice.
IGA6	IGA4			3	Use modern technology in performing engineering practice.
IGA1, IGA2	IGA1, IGA2	**l**	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	1	Capable of learning from and listening to co-members.
IGA3, IGA4	IGA3, IGA5			2	Shows leadership skills in managing a multidisciplinary group.
IGA5, IGA6	IGA4, IGA 5			3	Participates in discussions and comes up with sound suggestions to problems.
IGA6	IGA4	**m**	Understand at least one specialized field of electrical engineering practice	1	Demonstrate leadership and team management in a multidisciplinary group.
				2	Capable of learning from and listening to co-members
				3	Apply designed system or process to solve engineering problems

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes												
				a	b	c	d	e	f	g	h	i	J	k	l	m
FIRST YEAR	FIRST	3	Purposive Communication													
		3	Mathematics in the Modern World													
		3	The Life and Works of Rizal													
		3	Readings in Philippine History													
		3	Engineering Calculus 1	I				I								
		4	Chemistry for Engineers	I	I			I								
		1	Engineering Drawing and Plans				I			I						
		2	Movement Enhancement													
		3	National Service Training Program 1													
	SECOND	3	Understanding the Self													
		3	Ethics													
		3	Engineering Calculus 2	E				E								
		3	Engineering Data Analysis	I				I			I					
		4	Physics for Engineers	E	I		I	E								
		1	Computer-Aided Drafting						I	I		I		I		
		3	Computer Fundamentals and Programming		I							I		I		
		2	Fitness Exercises													
		3	National Service Training Program 2													
	OFF-SEM	3	Engineering Economics	E					I		I		I		I	
		3	Environmental Science and Engineering						E		E		E		E	
SECOND YEAR	FIRST	3	Contemporary World													
		3	Differential Equations	E				E								
		3	Electrical Circuits 1	E				E								
		1	Electrical Circuits 1 Lab		E									E		
		3	Engineering Mechanics	E				E								
		3	Fluid Mechanics	E				E								
		2	Basic Thermodynamics	E				E								
		3	Basic Occupational Safety and Health						E		E		E		E	
		3	Science and Technology and Society													
		2	Physical Activities Towards Health and Fitness I													
SECOND YEAR	SECOND	3	General Education Elective 1													
		3	Arts Appreciation													
		3	Electrical Circuits 2	E				E								
		1	Electrical Circuits 2 Lab		E									E		
		3	Advanced Engineering Mathematics for EE	E				E								I
		3	Electromagnetics 1	E				E								
		3	Electronic Circuits: Devices and Analysis	E				E								
		1	Electronic Circuits: Devices and Analysis Lab		E	E										
		2	Fundamentals of Deformable Bodies	E				E								
		2	Physical Activities Towards Health and Fitness II													

		3	General Education Elective 2														
		3	Electrical Machines 1					E		E							I
		1	Electrical Machines 1 Lab					E		E							
		2	EE Laws, Codes, and Professional Ethics						E	E	E				E	E	
		3	Feedback Control Systems	E				E		E				E			
		3	Material Science and Engineering						E	I	E						
		3	Industrial Electronics		E			E		I							
		1	Industrial Electronics Lab		E			E		I							
		3	General Education Elective 3														
		2	Logic Circuits Design and Switching Theory	E		E		E									
		1	Logic Circuits Design and Switching Theory Lab		E					E							
		3	Principles of Communication Systems	E				E									
		3	Engineering Management				E		E	E	E		E		E		
		3	Technopreneurship				E		E	E					E		
		3	Electrical Machines 2		E			E		E							
		1	Electrical Machines 2 Lab		E			E		E							
		1	Electrical Standards and Practice			E			D	E		E					
	SECOND	2	Instrumentation and Control		E					E							
		1	Instrumentation and Control Lab		E					E							
		2	Microprocessor Systems	E				E									
		1	Microprocessor Systems Lab		E	E				E				E			
		3	Methods of Research						E	E	E		E		E		
		3	EE Elective 1					E	E	D	D	D	E	E		E	
		2	Numerical Methods and Analysis	E				E						E			
		1	Numerical Methods and Analysis Lab	E				E						E			
		3	Capstone Design Project*	D	D	D		D	D	D	D	D	D	D		D	
		3	Power System Analysis			E		D		D			E	D			
		1	Power System Analysis Lab			E		D		D			E	D			
		1	Industrial Plant Visits and Seminars							D	D	D	E				
		3	Electrical Systems & Illumination Eng'g Design			D	E	D	E	D			E				
		2	Electrical Systems & Illumination Eng'g Design Lab			D	E	D	E	D			E				
		1	Fundamentals of Power Plant Engineering Design			E			E	E							
		2	Distribution Systems and Substation Design			D		E	E	E	E						
		1	Distribution Systems and Substation Design Lab			D		E	E	E	E						
		3	EE Elective 2					E	E	D	D	D	E	E		E	
		2	Electrical Apparatus and Devices		E			E		E							
		1	Electrical Apparatus and Devices Lab		E			E		E							
		3	Capstone Design Project*	D	D	D		D	D	D	D	D	D	D	D	D	
		2	On-the-Job Training (320 hrs)				D		D	D		D	D	D		D	
	SECOND	2	EE Elective 3: Correlation Course 1													E	
		2	EE Elective 4: Free Elective													E	

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

5 Very Likely
4 Less Likely
3 Cannot Tell
2 Not Likely
1 Definitely Not

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

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